NETFLIX AFTER HOURS

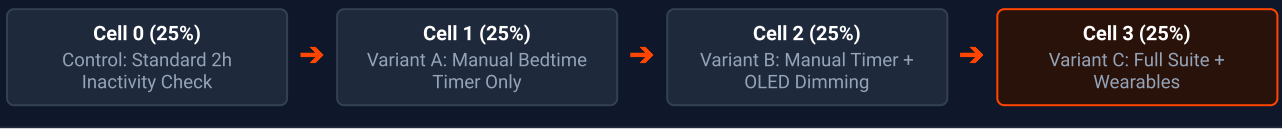
PATH B: A/B TESTING & FEATURE FLAG STRATEGY

Document Version: 1.0-FINAL
Experimentation Framework

1. EXPERIMENTAL FRAMEWORK & CELL ALLOCATION SCHEMATICS

Deploying late-night viewing controls across global accounts requires a multi-cell A/B testing strategy. Phase 1 targets 5% of global traffic, splitting users into four distinct treatment cells.

FIGURE 1.1: 4-CELL RANDOMIZED EXPERIMENT SPLIT FLOW



Experimental Cell	Assigned Capabilities	Traffic Allocation	Primary Evaluation Target
Cell 0 (Control)	Existing 2-hour inactivity prompt (Status quo)	25% of Test Vol	Baseline Egress & Rewind rate
Cell 1 (Variant A)	Manual Moon Icon + Preset Timers (15/30/45m)	25% of Test Vol	Manual Timer Adoption Rate
Cell 2 (Variant B)	Manual Timers + OLED Display Dimming Curve	25% of Test Vol	Luminescence Reduction Impact
Cell 3 (Variant C)	Full Suite: Dimming + Audio Decay + Wearable Sync + Recovery Card	25% of Test Vol	Full Bandwidth & Retention Delta

2. HYPOTHESES & METRIC EVALUATION MATRIX

> **Primary Hypothesis (H1):** Enabling dynamic display dimming and logarithmic audio decay during late-night hours (11 PM – 3 AM) will reduce non-interactive server egress bandwidth by ≥ 12% while increasing next-day episode resume accuracy by ≥ 20%.

Metric Category	Key Performance Indicator (KPI)	Success Target	Guardrail Threshold
Primary Business	Bandwidth Egress Volume (TB/hr)	**≥ 12% Reduction** (11 PM - 3 AM)	N/A
User Experience	Sleep Resume Accuracy Rate	**≥ 85%** Resume without rewind	Halt if < 70%
Monetization	Premium Tier Conversion Rate	**+1.5% Upgrade Delta**	Review if Net Negative
System Health	False Positive Cancellation Rate	**< 2.0%** Cancel sleep prompts	Alert if > 3.5%
Retention	7-Day Subscriber Retention Rate	**Net Neutral or Positive**	Halt if Stat-Sig Drop

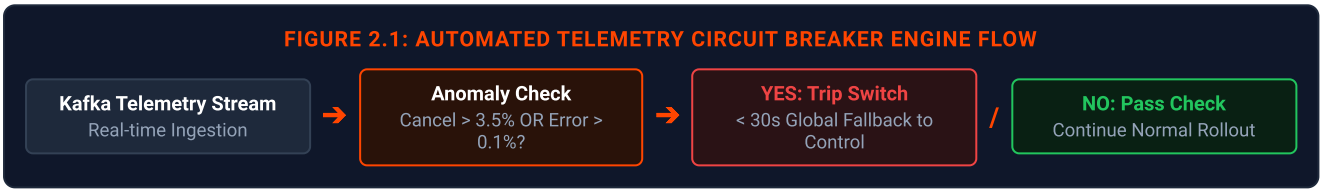
3. FEATURE FLAG RULE CONFIGURATION (LAUNCHDARKLY SCHEMA)

The feature flag payload (`netflix_after_hours_rollout_v1`) dynamically gates capabilities by local time window, plan entitlement, and experimental cell hashing.

```
{ "flag_key": "netflix_after_hours_rollout_v1", "enabled": true, "rules": [ { "rule_id":
"rule_time_window_check", "clause": "device_local_time BETWEEN '22:00' AND '04:00'", "action":
"ENABLE_AFTER_HOURS_PROMPT" }, { "rule_id": "rule_tier_entitlement_premium", "clause":
"user.subscription_tier EQUALS 'premium'", "capabilities": ["manual_timer", "oled_dimming", "audio_decay",
"wearable_auto_sync", "binaural_delta_layer"] }, { "rule_id": "rule_tier_entitlement_standard_ads",
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"sleep_recovery_card"], "restrictions": ["defer_ads_during_dimming"] } ], "experimentation": {
"cell_assignment_hash": "sha256(account_id + 'after_hours_v1')", "variant_distribution": { "control":
0.25, "variant_a": 0.25, "variant_b": 0.25, "variant_c": 0.25 } } }
```

4. AUTOMATED CIRCUIT BREAKER & KILL-SWITCH ARCHITECTURE

To protect live stream delivery, a real-time Kafka anomaly consumer monitors metric thresholds. If cancellation or error limits are breached, the circuit breaker executes a global kill-switch in under 30 seconds.



Fail-Safe Behavior Guarantee: If the circuit breaker trips (YES path), all active dimming or fading transitions immediately restore to 100% video brightness and standard audio output without interrupting video playback. If metrics remain within limits (NO path), the experiment proceeds through scheduled rollout stages.